

Program : Diploma in Wood and Paper Technology	
Course Code : 3122	Course Title: Wood Chemistry
Semester : 3	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L:2, T:1, P:0)	Periods per semester: 45

Course Objectives:

- To provide knowledge about the chemistry of raw materials in wood based industries
- To familiarize students about wood analysis techniques
- To impart knowledge about the manufacture of valuable products from wood other than paper.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Organic Chemistry Physical Chemistry		Applied Chemistry	1

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the chemistry, preparation and properties of organic compounds mainly carbohydrates.	11	Understanding
CO2	Propose the application of physical and nuclear chemistry in wood	9	Applying
CO3	Discuss the structure ,cell wall composition of wood and its chemical treatment	14	Applying
CO4	Illustrate the wood hydrolysis and destructive distillation of wood.	9	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4		3		3	2		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the chemistry, preparation and properties of organic compounds mainly carbohydrates.		
M1.01	Name the organic compounds using IUPAC nomenclature	3	Understanding
M1.02	Discuss the preparation, properties and uses of organic compounds	2	Understanding
M1.03	Explain the structure of cellulose, glucose and fructose	4	Understanding
M1.04	List wood extractive terpenes and their isolation from plants	2	Understanding
Contents: Organic Chemistry Classification of organic compounds - saturated and unsaturated compounds. IUPAC - nomenclature of simple organic compounds. Industrial methods of preparation, properties and uses of : Alcohol, phenol, aldehydes, ketones, carboxylic acid, amines, urea, poly - functional compounds - glycol, glycerol, Carbohydrates - classification of sugars, structure of glucose, fructose and cellulose, important reaction of these compounds. Terpenes - Essential oils, classification of terpenoids, Isolation, Isoprene rule and the examples of terpenes.			
CO2	Propose the application of physical & nuclear chemistry in wood.		
M2.01	Describe the theory of corrosion and effect of corrosion on wood	3	Understanding

M2.02	Analyse the colloids and their importance in paper making chemistry	3	Applying
M2.03	Apply the carbon dating to find the age of wood	3	Applying
	Series Test – I	1	
Contents: Physical Chemistry Corrosion-Theory of corrosion-dry & wet, methods to prevent corrosion. Degradation of wood by products of metal corrosion. Introduction to colloids, mechanical, electrical and optical properties of colloids. Gels & Emulsions; Importance of papermaking chemistry; Fiber-fiber water bonding; Rheology, surface energy, and surface tension of colloidal systems. Electro kinetic properties, Sorption and swelling of cellulosic materials. Nuclear Chemistry Natural and artificial radioactivity, radioisotopes and their uses, carbon dating in wood			
CO3	Discuss the structure, cell wall composition of wood and its chemical treatment		
M3.01	Describe the Anatomy of wood, difference between soft wood and hardwood	3	Understanding
M3.02	Explain cell wall structure and composition of wood	3	Understanding
M3.03	Summarize extractives and their effect on wood	3	Understanding
M3.04	Explain the chemical treatment and effect of chemicals on wood	5	Applying
Contents: Chemistry of Fibrous Raw Materials Structure of softwoods, hardwoods, and non-woods; Pulpwood species; Ultrastructure of cell wall, cell wall components - cellulose, hemi cellulose, lignin, extractives and inorganic components. Difference in cell wall composition of softwood and hard wood Wood extractives - Essential oils, Tannins, flavanoids and dyestuff, properties and uses. Effect of extractives on durability of wood, gluing characteristics, surface coating. Pectin - uses Chemical Treatment of wood - chemical resistance of softwood and hardwood. Effect of chemicals on wood - aqueous solutions of inorganic salts, alkaline degradation of wood, Oxidation and bleaching of wood. Action of water - effect of moisture content on strength & electrical conductivity of wood			

CO4	Illustrate the wood hydrolysis and destructive distillation of wood.		
M4.01	Explain wood analysis principles	3	Understanding
M4.02	Describe distillation of wood	3	Understanding
M4.03	Suggest different types of wood hydrolysis	3	Understanding
	Series Test – II	1	
Contents: Wood Analysis. - Preparation of sample - Moisture content-oven dry method and dean and stark Ash content, Cellulose - cross & bevan, alpha, beta, gamma, holocellulose Lignin, Extractives - steam distillation and solvent extraction (soxhlet extraction) Destructive distillation of wood - charcoal, Pyroligneous acid, wood tar. Wood hydrolysis - Acid hydrolysis, Bio - Chemical hydrolysis, enzymatic hydrolysis			

Text /Reference:

T/R	Book Title/Author
T1	R. H. Farmer, Chemistry in the utilization of wood, Elsevier, 2013
R1	Kocurek M. J., “Pulp and Paper Manufacture, Volume 1: Properties of Fibrous Raw Materials and their Preparation for Pulping (Ed. Kocurek M. J. and Stevens C. F.B.), TAPPI Press, 1983
R2	Franz F.P. Kollmann, Wilfred A.Jr. Cote, Principles of wood science & technology vol. I, Springer Berlin Heidelberg, 2012
R3	Browning B. L. “The Chemistry of Wood”, John Wiley & Sons. 1981
R4	Sjostrom E., “Wood Chemistry Fundamentals and Applications”, 2nd Ed., TAPPI Press. 1993
R5	K. S.Tewari, S. N Mehrotra, N. K. Vishnoi , “Textbook of organic chemistry” Vikas Publishing House Pvt Ltd.,
R6	B.R. Puri, L.R. Sharma, M.S. Pathania ,Physical chemistry, Vishal Publishing Company

Online Resources:

Sl. No	Website Link
1	http://www.ncert.nic.in/NCERTS/l/kech205.pdf /
2	http://ncert.nic.in/NCERTS/l/leph205.pdf